

day-78-spaceagent-bridge-integration

Day 78 - SpaceAgent + Bridge Integration

Status: PASS for developer-side Bridge gating

Blocker class: NONE for protected SpaceAgent research approval gating

Lane

SpaceAgent protected actions through Bridge approval.

What Was Implemented

- Wired protected SpaceAgent research requests through Bridge approval.
- Kept public read-only research packet planning unchanged and ungated when no protected boundary is requested.
- Kept browser execution, YouTube execution, Firecrawl execution, external writes, and protected actions disabled.
- Improved the SpaceAgent research classifier so private dashboard inspection is labeled as protected browser interaction instead of a vague no-research path.
- Added route tests proving protected research creates a scoped approval request and does not include the raw request body.

Files Changed

- 'src/app/api/gateway/space-agent/research/route.ts'
- 'src/lib/space-agent-research.ts'
- 'src/lib/space-agent-routes.test.ts'
- 'runtime/day-78-spaceagent-bridge-integration-proof.json'
- 'runtime/day-78-spaceagent-bridge-integration.md'
- 'runtime/day-78-spaceagent-bridge-integration.pdf'

Routes Changed

- 'POST /api/gateway/space-agent/research'

UI Behavior

No Mission Control or Gateway designer HTML, CSS, class names, or visual assets were changed. This lane only changes the backend truth for SpaceAgent protected research. Gateway and Agent Hub can now truthfully show:

- public read-only research packet planning stays non-executing
- protected browser/private-boundary research requires owner approval
- Bridge Session is required before protected browser work
- approval pending after protected request
- execution disabled
- writes disabled
- no fake completed research

Service Runtime Behavior

Runtime proof was performed against a short-lived local standalone Mission Control runtime on '127.0.0.1:3337'.

The proof used an in-process API key and did not write or modify any '.env' file.

Proof result:

- 'GET /api/bridge/space-agent/status' returned HTTP 200.
- Status still reported runtime blockers: Firecrawl missing credential and Playwright MCP service unreachable.
- 'POST /api/gateway/space-agent/research' with protected private-dashboard input returned HTTP 423.
- The protected request created a fresh pending Bridge approval request.

- The protected request included an audit event id.
- Secret-like owner input was redacted to '[redacted-secret]'.
- The protected request kept 'accepted_for_execution:false'.
- The protected request kept 'research_performed:false'.
- The protected request kept 'execution_enabled:false'.
- The protected request kept 'writes_enabled:false'.
- The protected request kept 'protected_actions_enabled:false'.
- A public read-only research packet request did not create approval and did not execute.

Proof artifact:

- 'runtime/day-78-spaceagent-bridge-integration-proof.json'

Tests Run

- 'pnpm exec vitest run src/lib/space-agent-routes.test.ts src/lib/space-agent-research.test.ts'
- 'pnpm run typecheck'
- 'pnpm run build'
- 'pnpm test'
- 'node scripts/protected-route-smoke-contract.mjs http://127.0.0.1:3337'
- 'git diff --check'
- 'node scripts/check-protected-file-invariants.mjs'
- 'node scripts/secret-scan-contract.mjs'
- 'node scripts/raw-exposure-scan-contract.mjs'
- '.env' diff check

Results:

- Focused SpaceAgent tests: 34 passed
- Typecheck: PASS
- Build: PASS
- Full test suite: 198 files passed, 1480 tests passed
- Protected route smoke: PASS, 41 routes checked
- Diff whitespace check: PASS
- Protected-file invariant scan: PASS
- Secret scan: PASS
- Raw exposure scan: PASS
- '.env' diff: clean

Deploy / Restart / Smoke Result

- No public exposure was added.
- No persistent production restart is claimed in this lane.
- A local standalone proof runtime was started on '127.0.0.1:3337', used for SpaceAgent proof and route smoke, then stopped.
- No listener remained on port 3337 after proof.

Remaining Blockers

Observed runtime blockers remain truthful and are not hidden:

- 'firecrawl_missing_credential'
- 'firecrawl_missing_credential; interactive_browser_actions_require_bridge_session'
- 'playwright_mcp_service_unreachable'

These do not block Day 78 developer-side Bridge gating because protected SpaceAgent research now creates an approval request and still does not execute.

Rollback

After commit:

```
""bash
git revert <day-78-spaceagent-bridge-commit>
```

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Commit / Push

Pending at report generation time.

Next Day

Day 79 starts next: OpenClaw+ + Gateway integration.